Customer Shopping Behavior Analysis

Analyzing **3,900 transactions** across product categories to uncover spending patterns, customer segments, and subscription behavior — guiding strategic business decisions.

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Dataset at a Glance

3,900 Rows

Transactional records across all product categories

18 Columns

Demographics, purchase details, and shopping behavior

37 Missing Values

Only in Review Rating — imputed by category median

Key Features

Age, Gender, Location, Subscription, Discount, Promo Code, Frequency, Shipping Type

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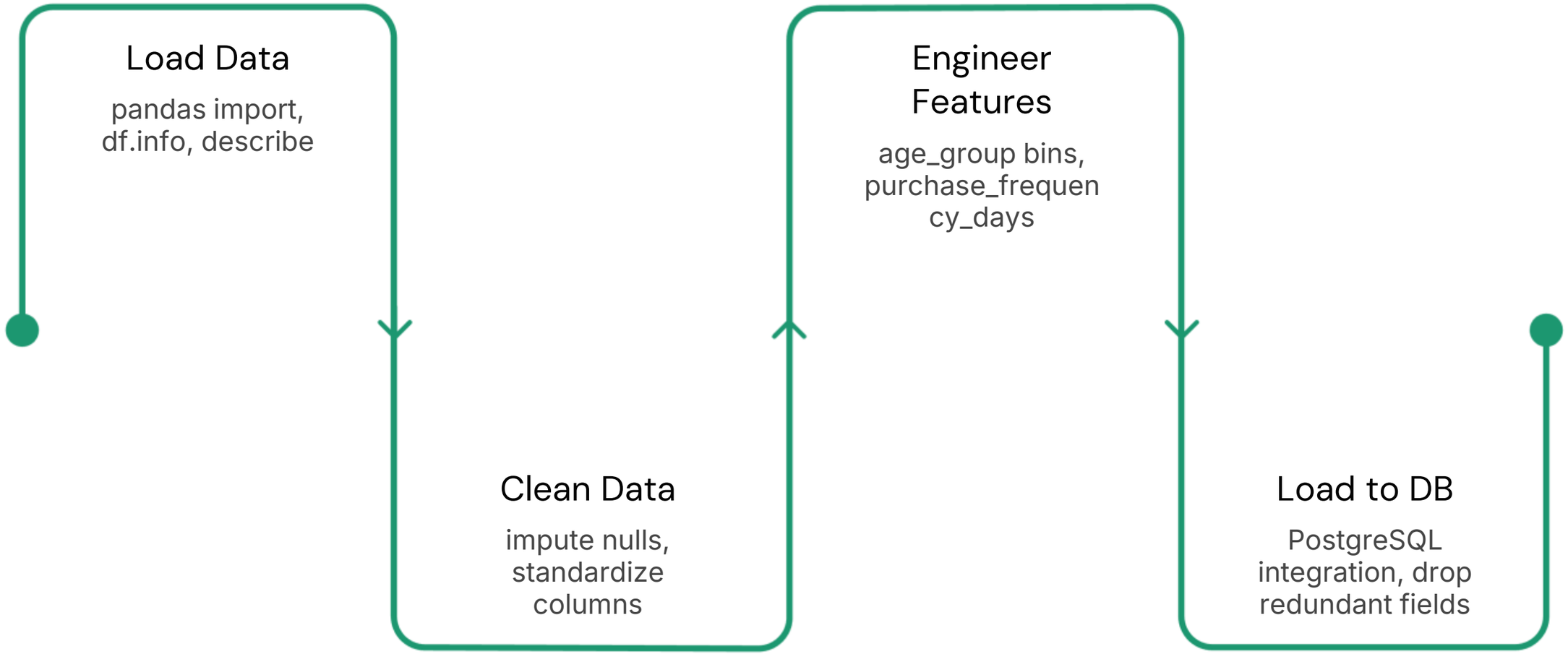
Exploratory Data Analysis in Python



Key steps included imputing missing Review Ratings by category median, creating **age_group** and **purchase_frequency_days** columns, and dropping the redundant **promo_code_used** field before loading into PostgreSQL.

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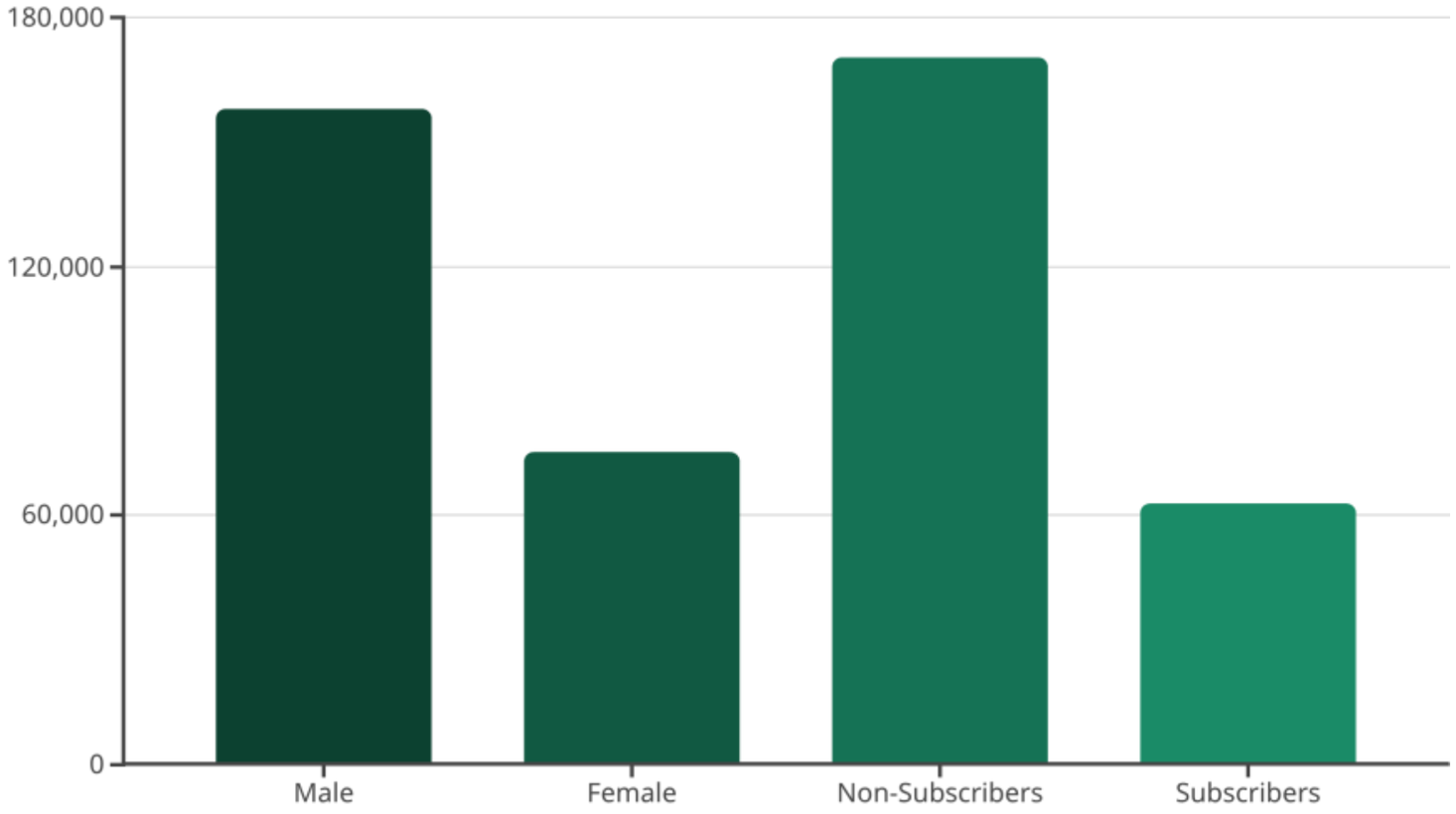
Revenue by Gender & Subscription Status



Key Findings

Male customers generate **2.1× more revenue** than females ($157,890 vs $75,191).

Despite being only **27%** of customers, subscribers spend nearly the same average per transaction (~$59.49 vs $59.87) — a strong upsell opportunity.

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Top Products by Rating & Discount Dependency

Top 5 Products by Average Rating

Item	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Top 5 Discount-Dependent Products

Item	% Discounted
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Hats, Sneakers, and Coats rely heavily on discounts — nearly half of all purchases use a promo. Balancing discount strategy with margin control is critical.

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Shipping & High-Spending Discount Users

Standard Shipping

Average purchase: **$58.46**

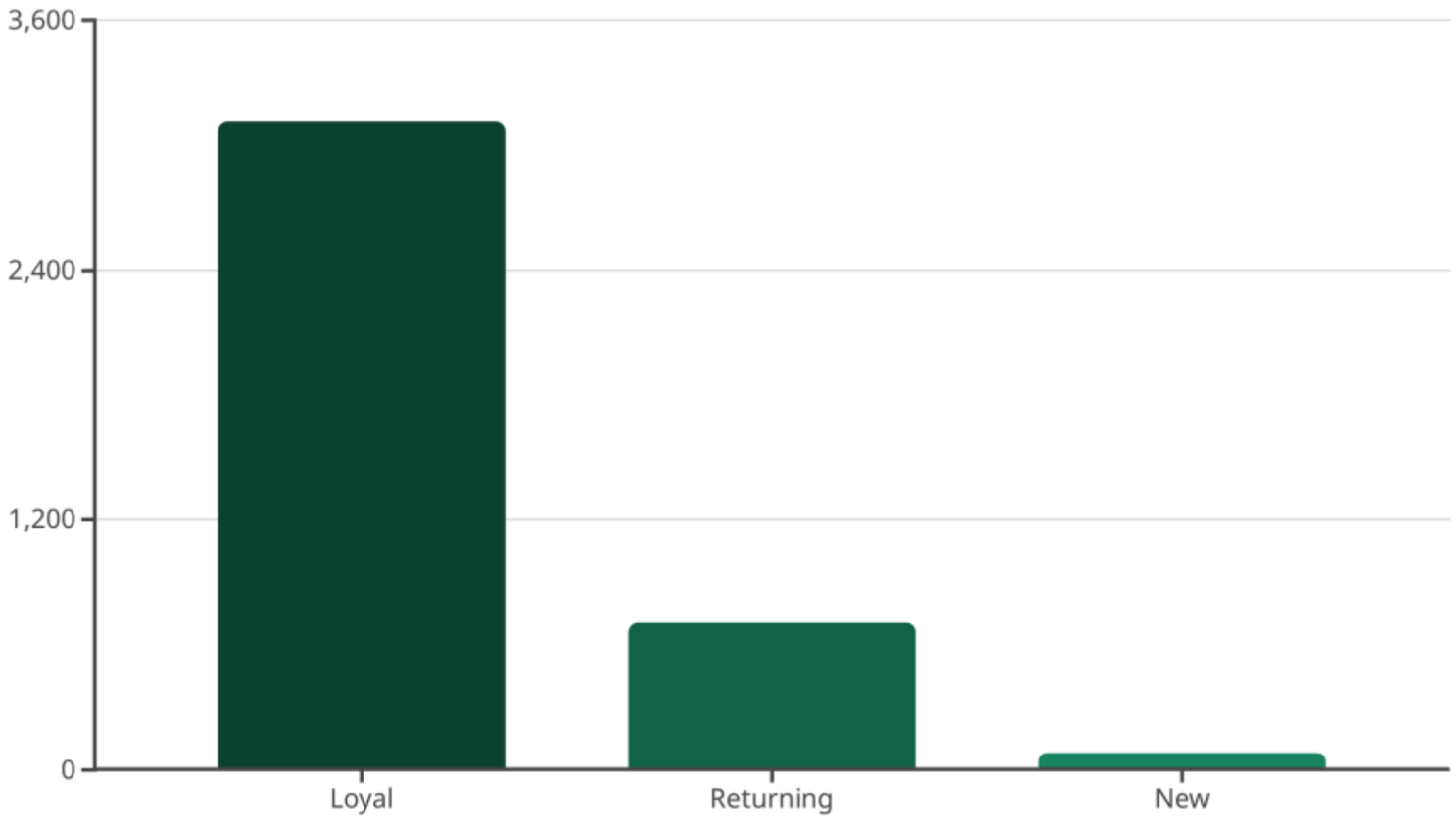
Express Shipping

Average purchase: **$60.48** — slightly higher spenders

839 High-Spend Discount Users

Customers who used discounts yet spent **above the average** purchase amount — prime targets for loyalty programs.

Customer Segmentation



Segment Breakdown

Customers classified as **New**, **Returning**, and **Loyal** based on purchase history.

Loyal customers dominate at 3,116 — but only 958 of repeat buyers (5+ purchases) are subscribers, revealing a major conversion gap.

Of 3,475 repeat buyers, only 958 are subscribed — a significant untapped opportunity.

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Revenue by Age Group & Top Products per Category

Revenue by Age Group (USD)

Age Group	Revenue
Young Adult	$62,143
Middle-aged	$59,197
Adult	$55,978
Senior	$55,763

Top Product per Category

- **Accessories:** Jewelry (171 orders)
- **Clothing:** Blouse (171 orders)
- **Footwear:** Sandals (160 orders)
- **Outerwear:** Jacket (163 orders)

Young Adults lead in revenue contribution. Blouse, Jewelry, Jacket, and Sandals are the standout performers across categories.

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Power BI Dashboard Highlights

3.9K

Total Customers

$59.76

Avg Purchase

3.75

Avg Rating

The interactive dashboard visualizes **Revenue by Category** (Clothing ~$100K leading), **Revenue by Age Group**, and **Subscription split** (73% non-subscribers vs 27% subscribers) — enabling real-time business exploration.

Business Recommendations

Boost Subscriptions

Promote exclusive benefits to convert the 2,518 unsubscribed repeat buyers.

Loyalty Programs

Reward returning customers to advance them into the "Loyal" segment.

Review Discount Policy

Balance sales boosts with margin control — 5 products show ~47–50% discount dependency.

Targeted Marketing

Focus on Young Adults and Express-shipping users; highlight top-rated products in campaigns.

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